

txt_to_img.ipynb

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Commands Code Text Run all

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[1]

29%

●

```
!pip install diffusers transformers accelerate torch torchvision
!pip install matplotlib
```

...

Requirement already satisfied: diffusers in /usr/local/lib/python3.12/dist-packages (0.38.0)

Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.12.0)

Requirement already satisfied: accelerate in /usr/local/lib/python3.12/dist-packages (1.14.0)

Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.11.0+cpu)

Requirement already satisfied: torchvision in /usr/local/lib/python3.12/dist-packages (0.26.0+cpu)

Requirement already satisfied: importlib_metadata in /usr/local/lib/python3.12/dist-packages (from diffusers) (8.7.1)

Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from diffusers) (3.29.3)

Requirement already satisfied: httpx<1.0.0 in /usr/local/lib/python3.12/dist-packages (from diffusers) (0.28.1)

Requirement already satisfied: huggingface-hub<2.0,>=0.34.0 in /usr/local/lib/python3.12/dist-packages (from diffusers) (1.19.0)

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from diffusers) (2.0.2)

Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from diffusers) (2025.11.3)

Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from diffusers) (2.32.4)

Requirement already satisfied: safetensors>=0.8.0-rc.0 in /usr/local/lib/python3.12/dist-packages (from diffusers) (0.8.0)

Requirement already satisfied: Pillow in /usr/local/lib/python3.12/dist-packages (from diffusers) (11.3.0)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.2)

Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)

Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)

Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from transformers) (0.25.1)

Requirement already satisfied: tqdm>=4.60 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)

Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate) (5.9.5)

Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)

Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)

Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)

Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)

Requirement already satisfied: avio in /usr/local/lib/python3.12/dist-packages (from httpx<1.0.0->diffusers) (4.13.0)

Variables

Terminal

Executing (57s)

Python 3



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```
Requirement already satisfied: pyparsing<2.5.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.0.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
```

```
[2]
✓ 48s
import torch
from diffusers import StableDiffusionPipeline
import matplotlib.pyplot as plt

# Use a simpler, more compatible model
model_id = "runwayml/stable-diffusion-v1-5"

print("Loading Stable Diffusion v1-5...")
```

```
... WARNING:torchao.kernel.intmm:Warning: Detected no triton, on systems without Triton certain kernels will not work
Flax classes are deprecated and will be removed in Diffusers v1.0.0. We recommend migrating to PyTorch classes or pinning your version of Dif
Flax classes are deprecated and will be removed in Diffusers v1.0.0. We recommend migrating to PyTorch classes or pinning your version of Dif
Loading Stable Diffusion v1-5...
```

```
[3]
try:
    pipe = StableDiffusionPipeline.from_pretrained(model_id)
    device = "cuda" if torch.cuda.is_available() else "cpu"
    pipe = pipe.to(device)
    print(f"Model loaded successfully on {device}!")
except Exception as e:
    print(f"Error: {e}")
```

```
... /usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:124: UserWarning:
The secret 'HF_TOKEN' does not exist in your Colab secrets.
```

{ Variables Terminal



Executing (1m 14s) Python 3

txt_to_img.ipynb

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Commands + Code + Text Run all

RAM Disk

[3] 2m

```
try:
    pipe = StableDiffusionPipeline.from_pretrained(model_id)
    device = "cuda" if torch.cuda.is_available() else "cpu"
    pipe = pipe.to(device)
    print(f"Model loaded successfully on {device}!")
except Exception as e:
    print(f"Error: {e}")
```

...

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:124: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set it as secret in
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.

warnings.warn(
model_index.json: 100% ██████████ 541/541 [00:00<00:00, 8.10kB/s]
Download complete: 100% ██████████ 5.48G/5.48G [02:50<00:00, 38.9MB/s]
Fetching 15 files: 100% ██████████ 15/15 [02:48<00:00, 12.45s/it]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher r
Loading pipeline components....: 100% ██████████ 7/7 [00:03<00:00, 3.16it/s]
Loading weights: 100% ██████████ 196/196 [00:00<00:00, 624.34it/s]
Loading weights: 100% ██████████ 396/396 [00:00<00:00, 942.45it/s]
Model loaded successfully on cpu!

Variables Terminal

Executing (32s) Python 3



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Commands + Code + Text ▶ Run all

RAM Disk

Loading weights: 100%  396/396 [00:00<00:00, 942.45it/s]
Model loaded successfully on cpu!

```
[4] ✓ 0s
def generate_simple(prompt):
    """Simple image generation function"""
    try:
        image = pipe(
            prompt,
            num_inference_steps=20,
            guidance_scale=7.5,
            width=512,
            height=512
        ).images[0]

        plt.figure(figsize=(8, 8))
        plt.imshow(image)
        plt.axis('off')
        plt.title(prompt)
        plt.show()
        return image

    except Exception as e:
        print(f"Generation error: {e}")
        return None
```

```
[5] ● # Test the simple version
```

Variables Terminal



Executing (43s) Python 3

txt_to_img.ipynb

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Commands + Code + Text Run all

RAM Disk

[4] ✓ 0s

```
plt.figure(figsize=(8, 8))
plt.imshow(image)
plt.axis('off')
plt.title(prompt)
plt.show()
return image

except Exception as e:
    print(f"Generation error: {e}")
    return None
```

[5]

Test the simple version
generate_simple("ninja")

5% 1/20 [01:32<29:17, 92.53s/it]

Variables Terminal

Executing (1m 36s) Python 3

txt_to_img.ipynb

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Commands + Code + Text Run all

RAM Disk

[4] 0s

return image

except Exception as e:

print(f"Generation error: {e}")

return None

[5] 15m

Test the simple version

generate_simple("ninja")

100% 20/20 [13:44<00:00, 38.22s/it]

ninja



Variables Terminal

5:02 PM Python 3